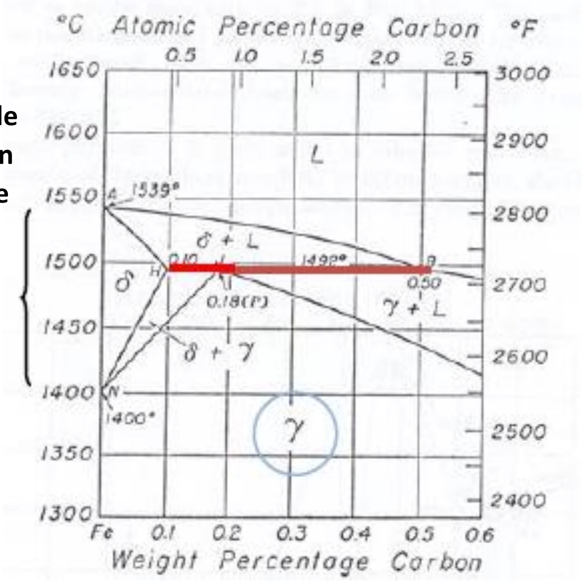
Peritectic transformation

It is important when we do fusion welding because the material in the welding region will first melt and then solidify according to the upper part of the diagram.

Lever rule applies in the same way



Invariant transformation:

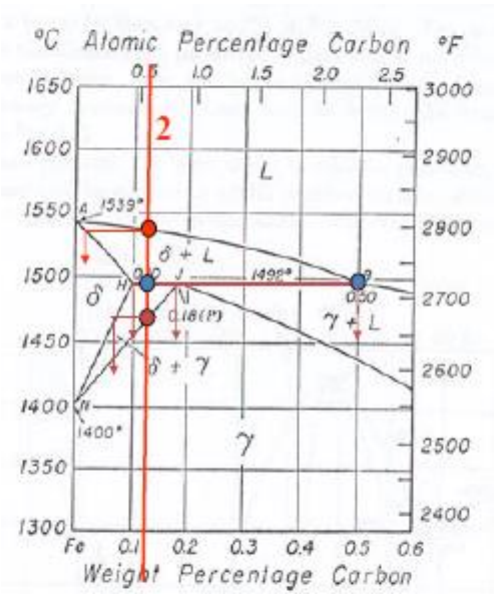
at T_{PER} 3 phases co-exist:

Solid δ: $\%C_H$

Solid γ: $\%C_P$

Liquid L: $\%C_B$

Each region has a different amount of C



Initial crystallization of δ, which composition changes together with that of the L as T ↓.

δ(H) starts to react with residual L(B) to give the new phase product γ(P)

T_{PER}= is also solidification temp.

Cooling.

The two phases δ and γ formed at T_{PER} proceed towards completion by diffusion processes; both phases will tend to the final $C\gamma = C_{orig.liq}$.

Exercise and solution

Given an alloy of 99.6 wt% Fe - 0.40 wt% C to a temperature just below A1 point, calculate:

a) weight fraction of Fe3C and ferrite (a).

b) the amount in grams of cementite formed per 100 g of alloy.

a) Use tie line (isotherm) A1

$C_\alpha = 0.022$ wt% C

$C_{Fe_3C} = 6.70$ wt% C

b) Using the lever rule over the A_1 tie-line

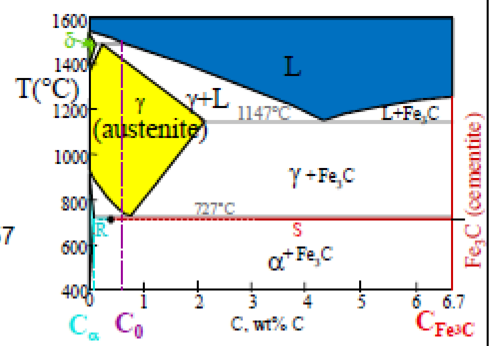
$$W_{Fe_3C} = \frac{R}{R+S} = \frac{C_0 - C_\alpha}{C_{Fe_3C} - C_\alpha}$$

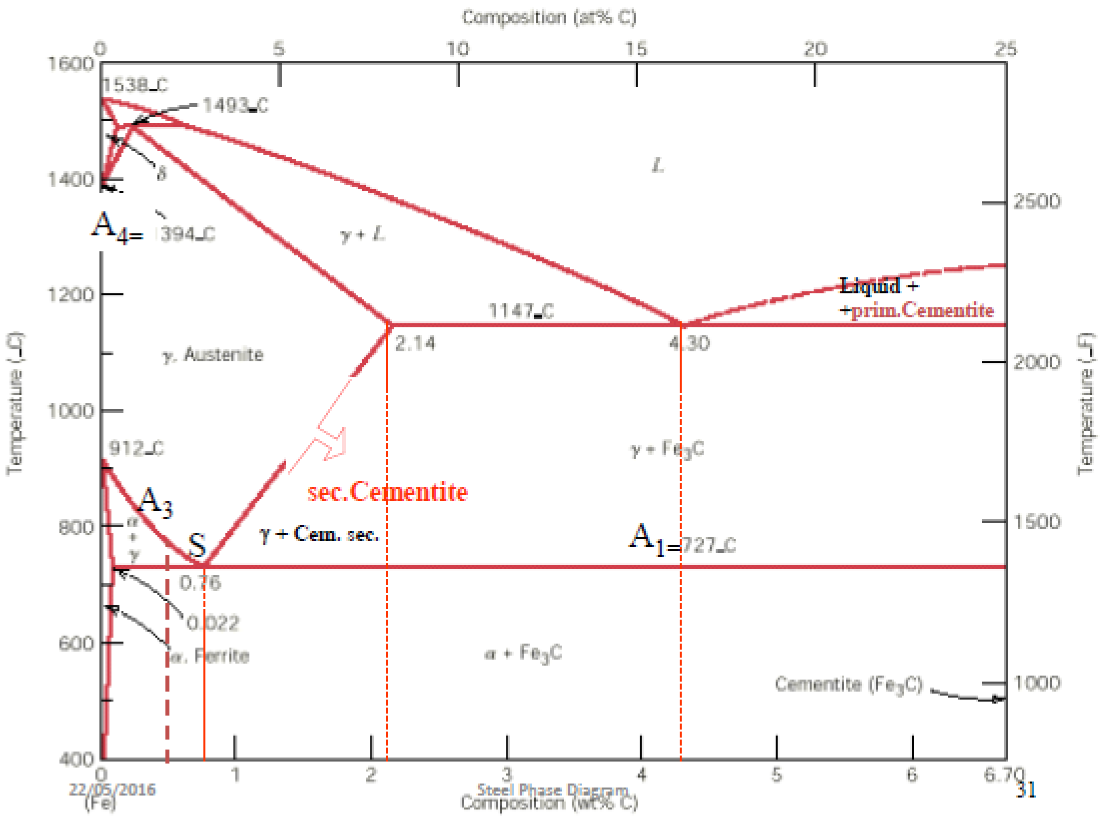
$$= \frac{0.40 - 0.022}{6.70 - 0.022} = 0.057$$

Amount of Fe_3C in 100 g

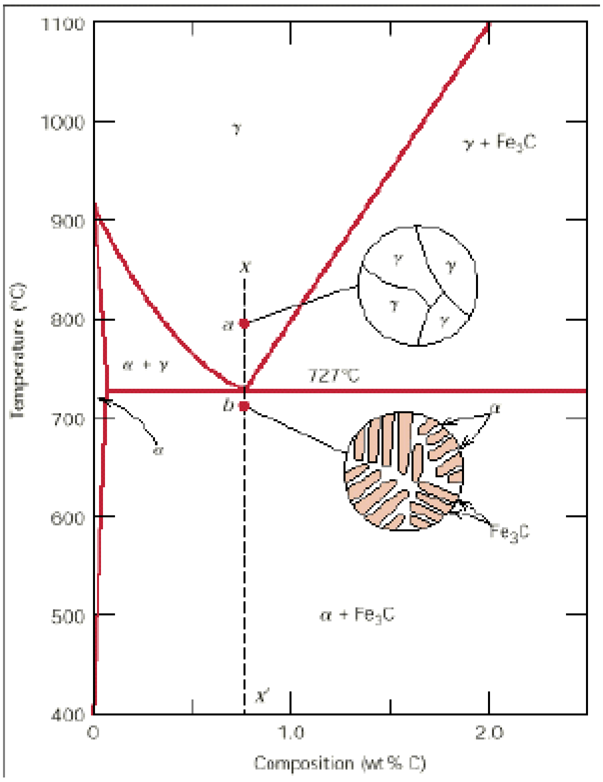
$= (100\text{ g})\ W_{Fe_3C}$

$= (100\text{ g})\ (0.057) \quad = 5.7\text{ g}$

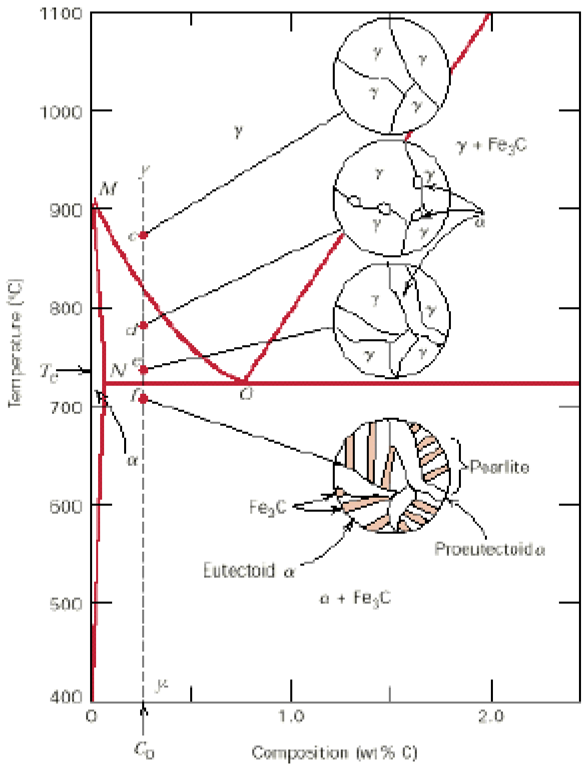


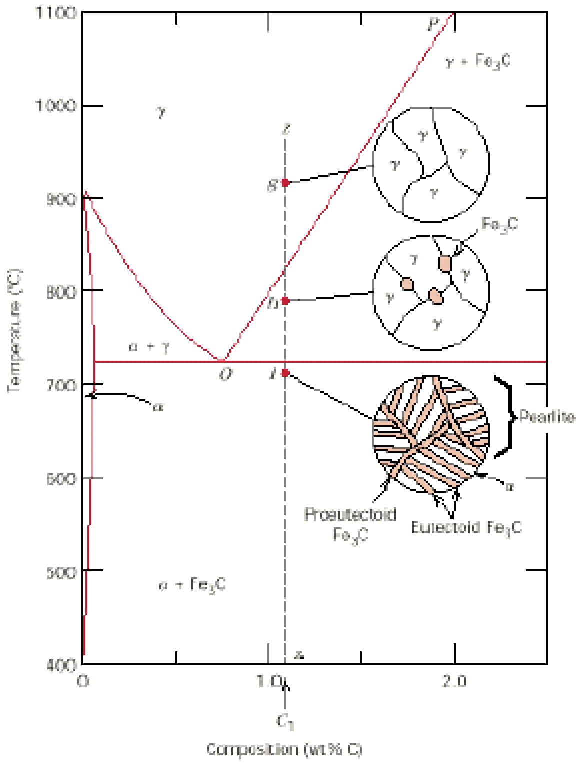


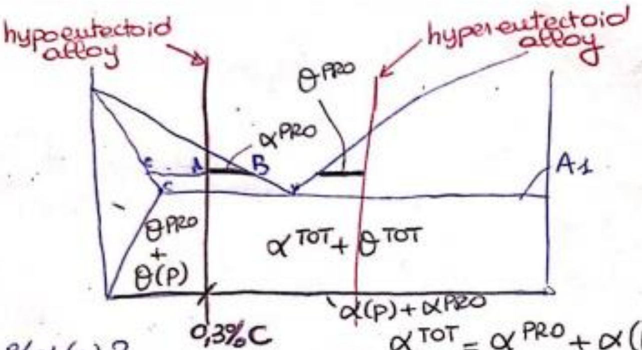
Eutectoid Steels



Hypo- and Hyper-eutectoid Steels







$\%\alpha(p)$?

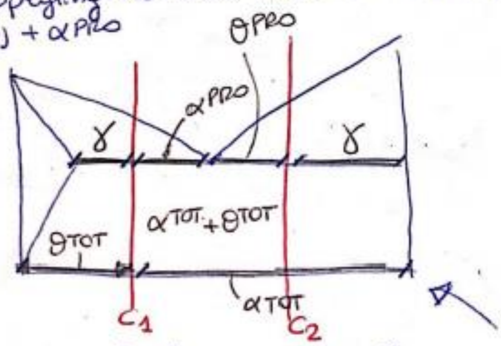
$\%\theta(p)$?

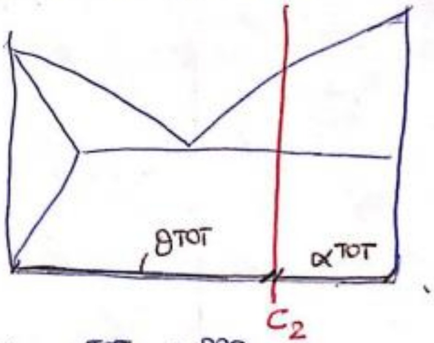
$\alpha^{TOT} = \alpha^{PRO} + \alpha(P)$

$\theta^{TOT} = \theta^{PRO} + \theta(P)$

By applying the lever rule at a T above A_1, we calculate α^{PRO} in the case of hypoeutectoid steels and θ^{PRO} in case of hypereutectoid steels

By applying the lever rule at room T, we calculate $\theta^{PRO} + \theta(P)$ and $\alpha(P) + \alpha^{PRO}$





If I want to calculate $\alpha(p)$ in $C_1 \rightarrow \alpha(p) = \alpha^{TOT} - \alpha^{PRO}$

$\theta(p) = \theta^{TOT} - \theta^{PRO}$

Lecture 10

Classification and designation of Steels – EN

Classification

Steels can be classified using different criteria:

- by alloying element content (ex. Carbon steel → a standard steel with C as its main alloying element)
- by heat treatment: steels for tempering, carburizing, nitriding, etc.
- by applications: structural, construction, tooling, etc.
- by degree of workability (formability): "automatic", cold forming, etc.

In the "automatic" applications, we have **machining** → **Turning**:

- Hard and brittle materials → small and fragmented chips (better!)
- Ductile materials → continuous chips

To increase the brittleness of the material and improve the automatic machining, we add a small amount of **Sulfur**. By adding **Lead**, instead, whenever the cutting edge reaches T equal to Tm of Lead, the chip will break down and we'll have a less troublesome machining. However, Lead is not used anymore because it's very toxic (it can easily go under the ground and pollute water).

Typically, standards do not follow only one criterion.

I. Classification – UNI EN 10020

If at least 1 element content > limits, the steel is named as **alloyed** (more expensive), otherwise is said **unalloyed**

Element	Limit, % (in mass)
Mn	1,65
Cr	0,3
Ni	0,3
Mo	0,06
W	0,3
V	0,1

Unalloyed steels (plain carbon steels):

- "quality"
- "special" -> with specifications: i.e. resilience, hardenability, purity (bounded inclusion content)

Alloyed steels:

- "quality": weldable with small grain size, for railways, for sheet being cold formed with deep drawing ability, for magnetic use, etc.
- "special" -> others: tempering steels or carburizing, for tools, rapid steels, etc.

Stainless Steel (inox): Cr > 10.5%, C < 1.2%

II. Alphanumeric Designation

Designation: a code that we use to identify very quickly a steel

- **a.** Based on **mechanical/physical properties and uses**
- b. Based on **chemical composition**

II. Alphanumeric Designation based on mechanical/physical properties and uses

[use, group*] ([subgroup]) [property value**] ([additional symbols***])

Example:

S235JR

=

Structural use, **Rp02 235 MPa**, **Resilience 27**

* Application of the steel
** Yield or Tensile stress
*** not always used, provided by the supplier (Charpy impact test)

Ductile to brittle transition → materials which are ductile at room T become brittle at lower T. To stress the toughness of a material, we have to specify the amount of energy absorbed by impact and the temperature at which the energy must be absorbed.

In some alloys, it is not always possible to specify the yield stress because the linear (elastic) part and the plastic part don't have a well-defined connection point → we have to use 0.2% plastic strain yielding (Proportionality Limit).

Use groups (a few):
S, structural
P, high pressure
R, railway
D, deep drawing sheets
H, high strength deep drawing sheets
M, magnetic properties
…

<u>Group S: Structural steels</u>
<u>Weldable beams, UNI-EN-10025</u>

These are very commonly used steels, cheap and very easy to produce.
In the old designation, the fracture stress is considered. In the present one, the yield stress is considered.

old	Prssent	R_{p02} [MPa]	R_m [MPa]	ε_{min} [%]	C	Mn	Si	S,P
Fe 360	S235	235	360-510	20	<0,17	<1,4	-	<0,035
Fe 510	S355	355	510-680	15	<0,2	<1,6	<0,55	<0,035

R_{p02}: Proportionality limit
R_m: Fracture stress
ε_{min}: Ductility expressed as a percentage

S235	Diameter [mm]	<100	100-150	150-250
	R_m [MPa]	360-510	350-500	340-490

Different strength by varying the size → R_m decreases as the diameter increases.

- ✓ The yield stress and ultimate stress are prescribed and can be provided by the producer by various combinations of compositions and thermomechanical treatments.
- ✓ The max limits for C, Mn are prescribed for welding purposes.

400 yield stress steel is as safe as S235.
Ductility is lower as the stress is higher.
Usually when strength increases, the toughness decreases. With grain refinement, we get high toughness and high strength. Toughness is required in dynamic loads (ex. Vibrations).

<u>Group S: Structural steels</u>
<u>Additional symbols</u>

Examples:
S235JR: resilience at least 27 J at temperature of +20 °C
S235J2: resilience at least 27 J at temperature of -20 °C
→ More severe toughness requirement because energy must be provided at low temperature

In general: S235XY

X is one of:
- J: > 27J
- K: > 40J
- L: > 60J

Y is one of:
- R: 20°C (room temperature)
- 0: 0°C
- 2: -20°C
- 3: -30°C

S550GD +Z, +ZF, +ZA, +AZ construction steel hot-dip Zn-coated

Group D: deep drawing sheets

Low carbon steels for cold forming

Subgroup (D = Drawing):

- D**X** – nor specified
- D**C** – cold rolled
- D**D** – hot rolled – **UNI-EN-10111:**

old	present	R_{p02} [MPa]	R_m [MPa]	A [%]	C	Mn
Fe P11	DD11	170-340	440	29	<0,12	<0,6
Fe P13	DD13	170-310	400	29	<0,8	<0,4

(11, 13: conventional numbers, they have no particular meaning)

- ✓ The yield stress and ultimate stress are prescribed and can be provided by the producer by various combinations of compositions and thermomechanical treatments
- ✓ The max limits for C, Mn are prescribed for welding purposes

DX57D +Z, +ZF, +ZA, +AS Hot-dip Zn-coated cold forming steel

Group H: High strength deep drawing sheets

UNI-EN-10346

group	subgroup Rolling type	property R_m [MPa]	Additional symbols Microstructure
H	C - cold	T(n)nnn	X - dual phase
			F - ferritic-bainitic
	D - hot		T - TRIP
			C - complex phase
			M - martensitic

grade	R_{p02} [MPa]	R_m [MPa]	A [%]	C	Mn
HCT600X	340 - 420	> 600	> 20	< 0.17	< 2.2
HDT950C	720 – 920	> 950	> 9	< 0.25	< 2.2
HDT1200M	900 - 1150	> 1200	> 5	< 0.25	< 2

Martensite increases the tensile strength, but it depresses the elongation to min 5.

Fracture stress taken into consideration instead of yield stress, because this type of steels is used to make safety parts (pillars, beams in cars, …) → resistance to impact is required: we need fracture stress.

II. Alphanumeric Designation based on chemical composition

Unalloyed steels with Mn < 1%:

CNN

NN = [nominal % C] x 100

Es.: **C45**, **C10**

Unalloyed steels with Mn>1% or alloying elements with each < 5% (low alloy steels)

NN [El_1][El_2]...[El_j] [n_1]-[n_2]-...[nj]

NN = [nominal % C] x 100

El_i: symbols of alloy elements

n_i: [nominal % element] x factor

NOT all elements appear in the grade notation

i.e., Mn usually is not prescribed when < 1%

Ex.: **28Mn6**, **42CrMoS4**, **20NiCrMo2**

Elements	factor	
Cr,Ni,Mn,Co,Si,W	4	More common
Al,Mo, ...	10	
S, ...	100	
B	1000	

Alloyed with at least one element with > 5% (except for rapid steels):

X NN [El_1][El_2]...[El_j] [%1]-[%2]-...[%j]

NN = [nominal % C] x 100

El_i: Symbols of alloying elements

%i: nominal alloy element content

Ex.: **X100CrMoV5**, **X38CrMoV5-3**

Rapid steels (**H**igh speed **S**teels, **HS**)

HS [% W]-[% Mo]-[% V]-[% Co]

% …: nominal content of the element,

Es.: **HS 6-5-3-8**

HS are used for cutting tools, for making dies, punch, knives, hammer, … These tools must be really hard, so they require many complex carbides.

V is the alloy element which will produce very fine and very hard carbides.

Tempering steels – UNI EN 10083

	S	C	Mn	Cr	Mo	P
C45E	<0,035	0,42-0,5	0,5-0,8	-	-	<0,035
C45R	0,02-0,8					
42CrMo4	<0,035	0,38-0,45	0,6-0,9	0,9-1,2	0,15-0,3	<0,035
42CrMoS4	0,02-0,8					

Carburizing steels – UNI EN 10084

	S	C	Mn	Cr	Ni	Mo
C10E	<0,035	0,07-0,13	0,3-0,6	-	-	-
C10R	0,02-0,04					
20NiCrMo2-2	<0,035	0,17-0,23	0,65-0,95	0,35-0,7	0,4-0,7	0,15-0,25
20NiCrMoS2-2	0,02-0,04					

Tool steels – UNI EN ISO 4957

	C	Mn	Cr	Mo	V	Si
X100CrMoV5	0,95–1,05	0,2–0,6	4,8-5,5	0,9-1,2	0,15-0,35	0,1-0,4
X38CrMoV5-3	0,35-0,4	0,3–0,5	2,7-3,2	2,7-3,2	0,4–0,6	0,3–0,5

III. Alphanumeric Designation – UNI EN 10027

1. XXYY

1 = "steels", XX = group, YY = progressive number

"00" base steels
S253JR → 1.0037
S355JR → 1.0045

"03" steels with C < 0.12%
DD11 → 1.0332
DD12 → 1.0335

"09"
HCT600X → 1.0941
HDT950C → 1.0958
HDT1200M → 1.0965

"11" e "12" Carbon steels
For structural/mechanical uses
C45E → 1.1191
C45R → 1.1201
C10E → 1.1121
C10R → 1.1207

"72" steels with Cr and Mo, Mo ≤ 0.35%
42CrMo4 → 1.7225
42CrMoS4 → 1.7227

"65" steels with Cr, Ni and Mo
20NiCrMo2-2 → 1.6523
20NiCrMoS2-2 → 1.6526

IV. American designation AISI/SAE

Ex: tempering steels

[group][%C x 100]

1040

Approximate correspondence

	Number	% C	% Mn	% Si	% Ni	% Cr	Others
	1020	0.18–0.23	0.30–0.60				
C45E	1040	0.37–0.44	0.60–0.90				
	1060	0.55–0.65	0.60–0.90				
	1080	0.75–0.88	0.60–0.90				
	1095	0.90–1.03	0.30–0.50				
	1140	0.37–0.44	0.70–1.00				0.08–0.13% S
42CrMo4	4140	0.38–0.43	0.75–1.00	0.15–0.30		0.80–1.10	0.15–0.25% Mo
	4340	0.38–0.43	0.60–0.80	0.15–0.30	1.65–2.00	0.70–0.90	0.20–0.300% Mo
	4620	0.17–0.22	0.45–0.65	0.15–0.30	1.65–2.00		0.20–0.30% Mo

AISI/SAE Steel Nomenclature

I don't have these slides

Lecture 11

Modern production of steels

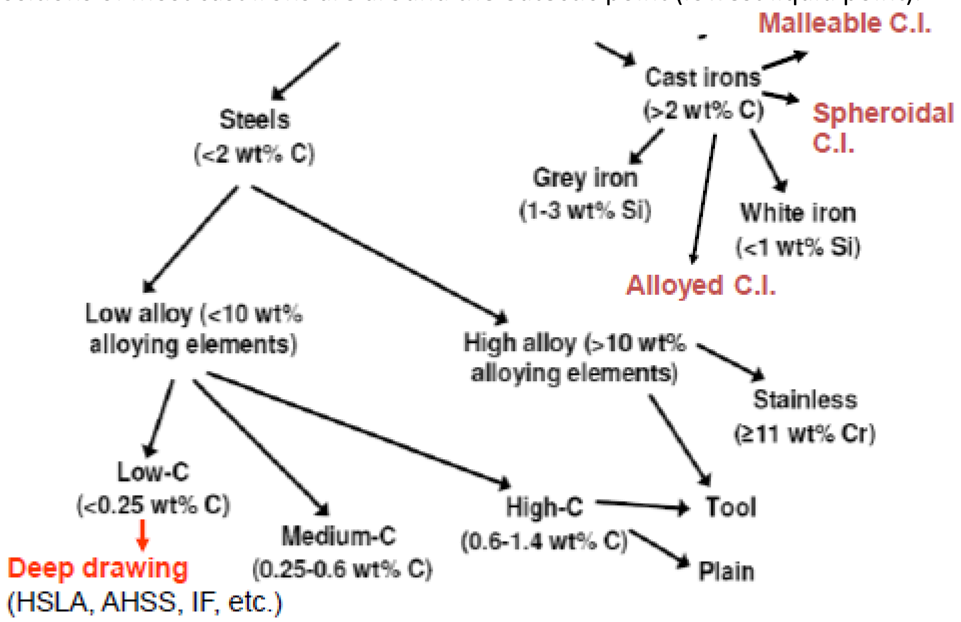
Relationship between microstructure, properties and process.

Classification

%C < 0.8 → Steels

%C > 2% → Cast Iron: main alloying elements are C and Si

- Low melting temperature.
- Compositions of most cast irons are around the eutectic point (lowest liquid point).



Cast irons tend to be brittle, except for **malleable**.

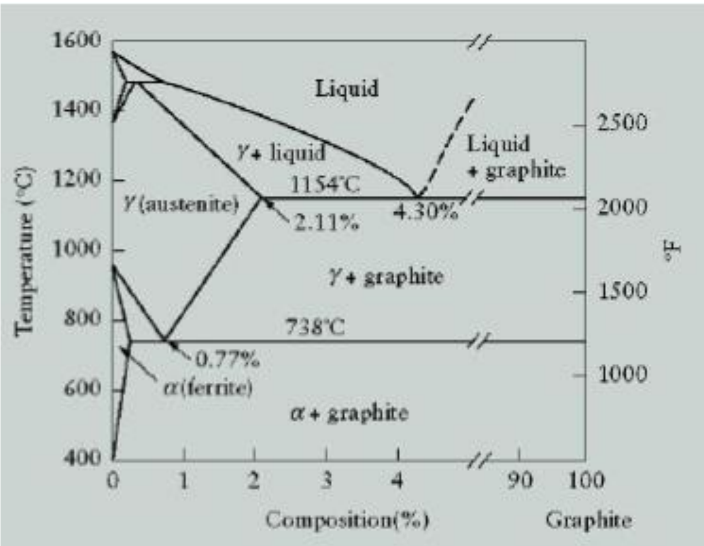
Spheroidal cast iron can be subjected to high temperature like medium steel following the same rules.

White iron is not very easy to produce; it has high resistance to corrosion.

Medium Carbon steels are used for tempering → steels are quenched and tempered in order to achieve high strength.

Equilibrium Fe-C Phase diagram

Steels cannot be produced from the liquid state the temperature is very high. The easiest way is to use the lowest liquid point in the diagram → **Eutectic point**.



Production of Steels

1. Integral production (minerals):

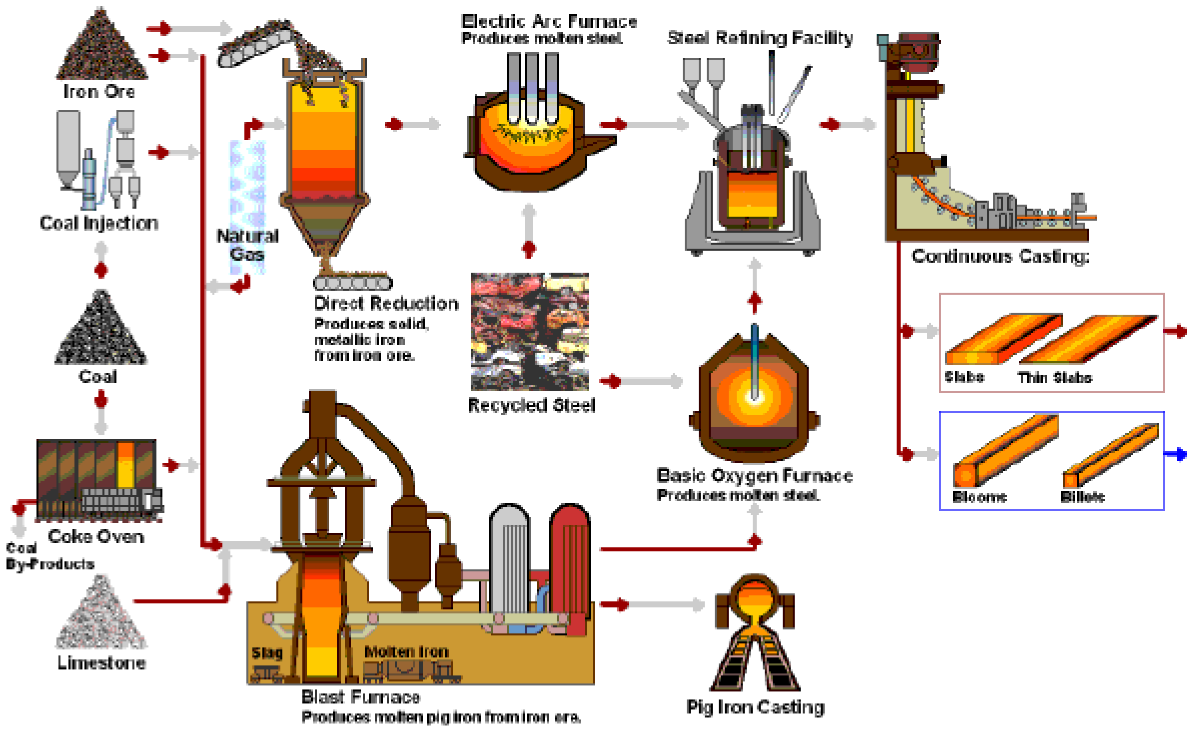
- Blast furnace production → pig iron (liquid produced) → Cast iron
- BOF-converter (to convert cast iron into steel) + pig iron → Steels

2. Secondary production (scraps + pig iron): BOF (Basic Oxygen Furnace) → Steel

3. Direct production: electric furnace (scraps) → Alloy Steels (Special steels)

Starting from steel we get steel. This type of production is dominant because it's very fast and flexible (we can control the composition of scraps in order to get the required steels).

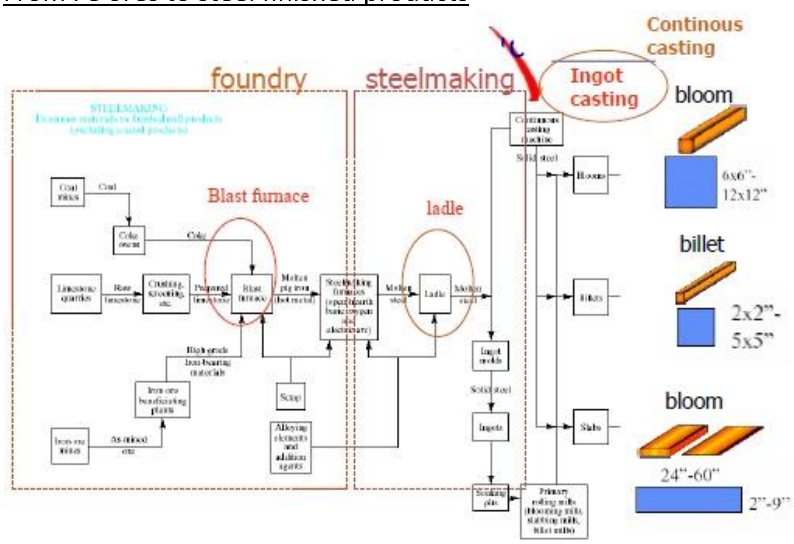
Integral Production of Steels



Carbon Coke reduces oxides from iron.

Iron Ore is the primary raw material. Putting **Carbon Coke** and **Limestone** (like Calcium carbonate, Calcium oxides, …) into the mixture conventionally prepared before filling the **Blast Furnace** will create many eutectic points → this will facilitate the melting of Iron Ore which has a very high melting point.

From Fe ores to steel finished products

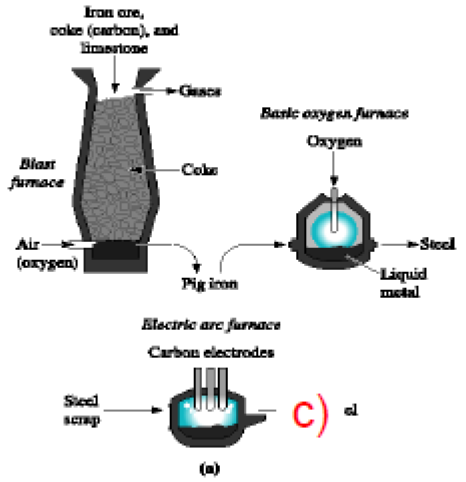


Foundry: for cast iron production

Steelmaking: for steel production

Ladle: container which is usually filled with the liquid to make chemical composition arrangement.

<u>Fundamental steps</u>

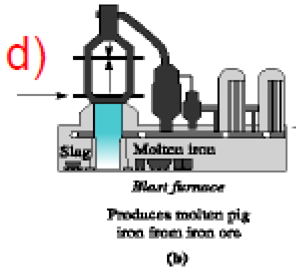


a) The raw materials for steelmaking (iron ore, limestone and carbon coke, oxygen) are converted in the blast furnace to molten iron (PIG IRON at high concentration of impurities: S, P, gases) containing about 4.3% carbon at a rate which varies up to about 8 million tons per annum.

b) The high content of C is reduced by injecting oxygen-argon to pig iron in a special furnace - basic oxygen furnace (BOF) – to produce liquid steel with the correct %C.

c) Alternatively, we can use an electric furnace (arc furnace or induction furnace) to produce liquid steel starting from selected iron scraps (direct process).

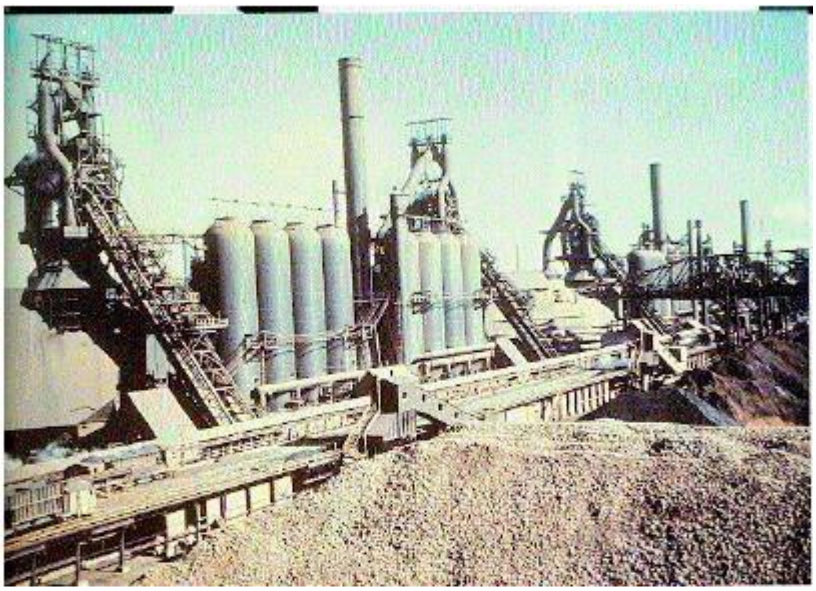
d) Scheme of the overall plant;



C-coke = sponge-like carbon produced from coal to expel the organic matter and gases

Limestone = calcium carbonate - is added as a flux for easier melting and slag formation.

<u>Iron ores</u>



Raw material covered by water jets to avoid pollution

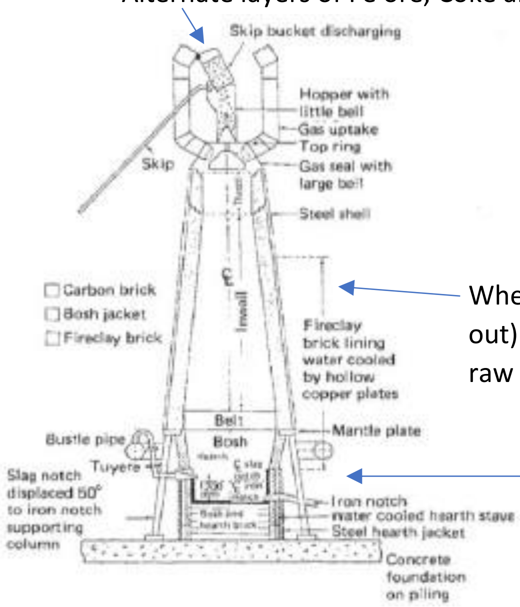
Blast Furnace operations

Transfer of pig iron to steelmaking workshop

Selected Fe scraps for BOF charge

The Blast Furnace

Alternate layers of Fe ore, Coke and Limestone



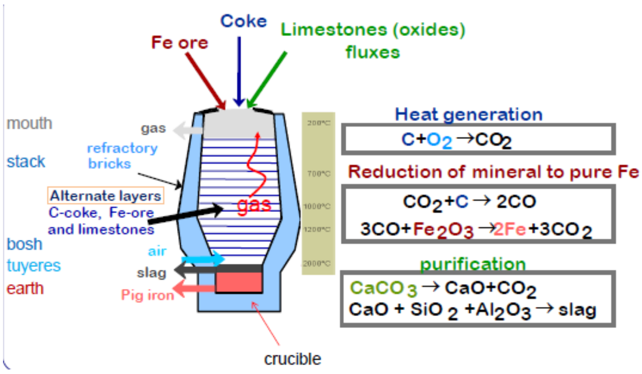
(The integral process by the blast furnace is by far the most economic process for producing steel, but economies which have not the capacity to utilize steel production of this magnitude may utilize the direct reduction of iron ore to sponge and powder instead).

Where combustion occurs (CO_2 produced and released out) with a resulting increase in temperature → melting of raw materials

Oxides that work as filters for liquid Fe before it collects on the bottom part and then it will be extracted from the bottom to another station (production of pig iron)

Reactions in the Blast Furnace

Gas interacts with the minerals used.



O_2 is injected from the outside.

CO is the real reducing agent.

Fe_2O_3 → Ferric Oxide

$CaCO_3$ → Calcium Carbonate

Purification: all the products are oxides which are very light (lower density than Fe) → they will float on the surface of the liquid.
Slag behaves like a filter: when the upper layers of Fe will be molten and introduced, they will be constrained to go through this slag → Fe is purified, we have pure Fe at the bottom. At slag lever, the temperature is very high to prevent Fe from solidifying.

Cast Iron production

1 t cast iron = { 600 kg coke; 2000 m³ air ⇒ 3000 m³ hot gases; 0.5 t slag }

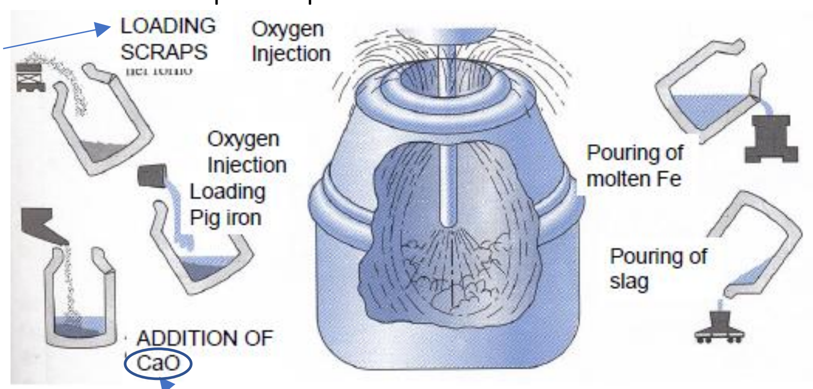
From combustion

3000 m³ hot gases ⇓ 1/3 used for preheating O_2.

Decarburization: BOF converter by oxygen injection launce

The decarburization is done in order to convert cast iron into steel.
Molten pig iron from the Blast Furnace is poured into the ladle, a large refractory lined container. The metal on the ladle is sent directly to BOF, High purity oxygen at high pressure is introduced at supersonic speed into the surface of pig iron.
Pre-treatment of the Blast Furnace pig iron is done externally to reduce S and P before charging pig iron into the converter → Mn is added and Sulphur impurities are reduced to MnS.

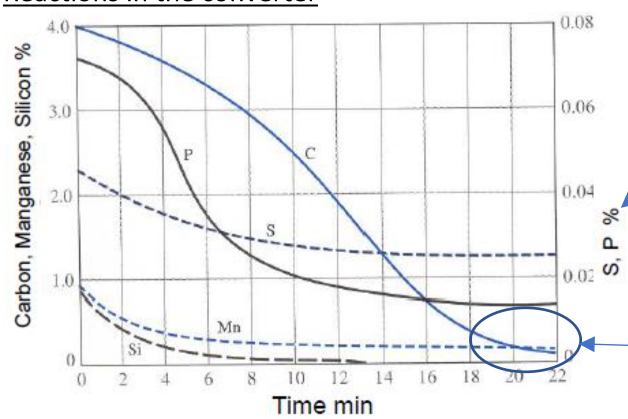
They will collect at the bottom



Oxygen introduced at high pressure
↓
Reactions
↓
High T
↓
Carbon will escape as CO or CO_2 from the top

Refinement agent to purify the liquid

Reactions in the converter



As mentioned before, Mn contributes to purification, it's always present.

Very low

Low Carbon Steels

Dangerous elements: P, Cu, Sn, Pb

Max tolerable content of impurities in high quality steels:

S= 0.03%
P= 0.02%
Cu=0.3%
Sn=0.06%
Ni=0.3%
Cr=0.2%

NB. S and P are dangerous elements not only for plain carbon steels but for all steel grades except for rephosphorized and resulphurized steels in which a deliberate amount of these element is added.

Si and **Mn** are always present because contribute to purification processes.

S and P in steels

Plain carbon steels:

- C < 2 % (typically <1%)
- Si = 0.2 - 0.3 %
- Mn = 0.3 - (< 0.7 %)

Alloy steels:

- Cr, Ni, Mo, V, Mn, S (as alloying element): >1%
- Impurities (P, S): (required< 0.05 %)
- P: embrittlement due to segregation at gbs of Fe_3P,
- S: hot embrittlement due to segregation of FeS

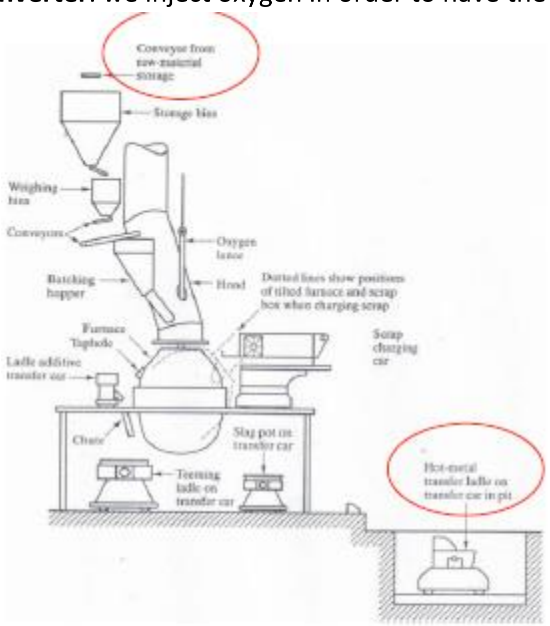
High speed steels (Numerically controlled machining):

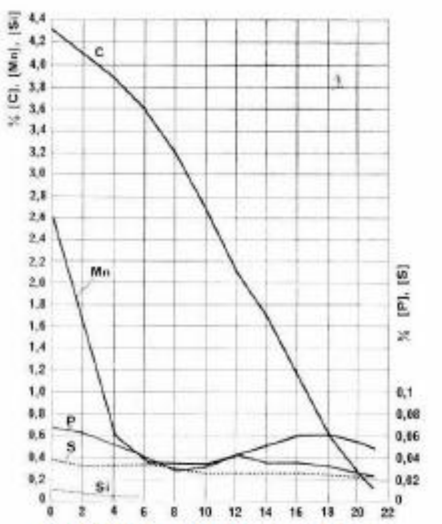
- S < 0.06-0.35 %
- S: makes the chips brittle, i.e. easier removal of chips

BOF Operations (for primary and secondary processes)

BOF can operate as a furnace and as a converter:

- **Furnace**: we put gasoline to make combustion occur → melting of scraps mixed with pig iron
- **Converter**: we inject oxygen in order to have the decarburization of the liquid

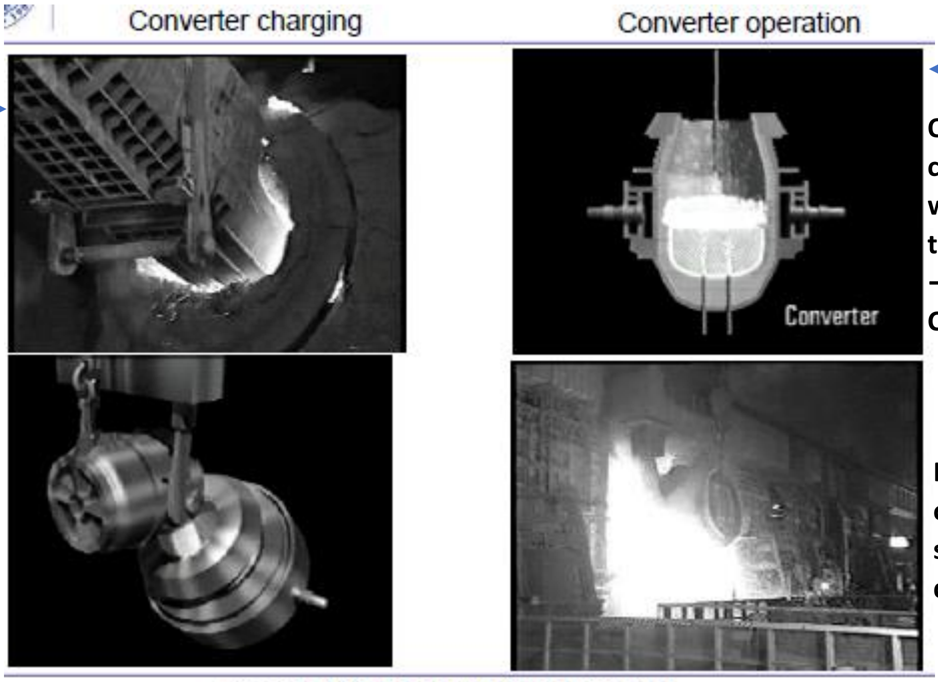




Purifications reactions idue to O_2 injection vs injection time

Converter Operations

Pig iron into converter



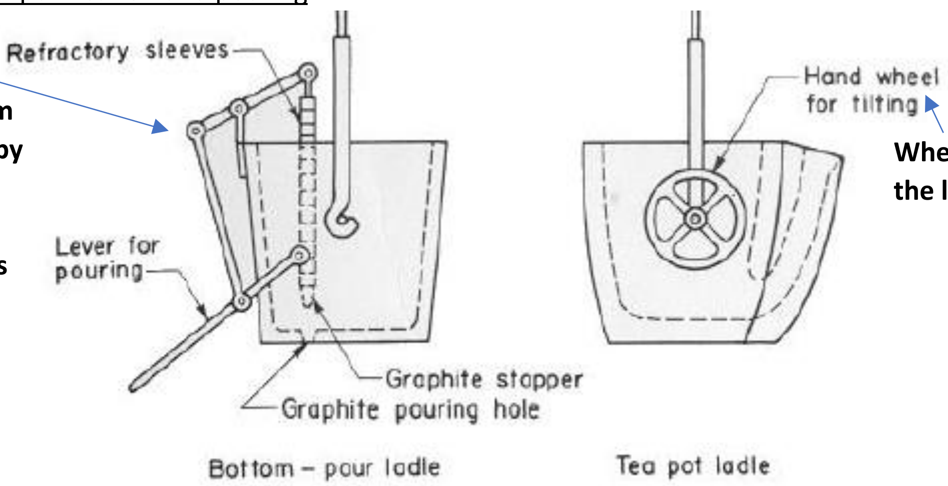
Charge of pig iron into the converter

O_2 will make combustion with C inside the pig iron → C out as CO or CO_2

Reaction occurs → sparkle → dangerous

Ladles: for post-purification and pouring

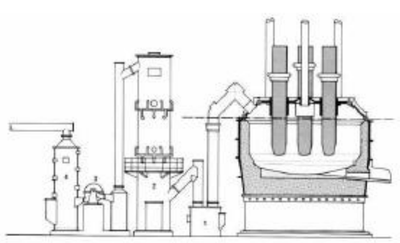
Liquid escaped from the bottom by rising the lever which opens/closes the hole

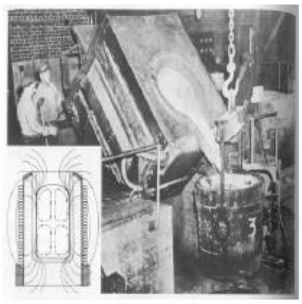


Wheel to rotate the ladle

Figure 11-60. Types of ladles for use in pouring castings.

Secondary Processing (EAF, Induction furnace)


Electric arc furnace for melting and reduction. Left: plant to recover energy from hot gases.


Induction furnace during pouring into ladle. Left: fundamental principle of electromagnetic Induced currents used to melt the metallic charge

EAF is very fast because after we charge the furnace, an arc will be formed between the graphite electrodes and the metal to be melted (electric current is passed through the electrodes to form the arc). The arc will melt the scraps.

In the **induction furnace**, instead of the arc we can use a coil to melt the scraps. After passing current to the coil, the magnetic field generated will penetrate the metal and will heat it till it melts.

Secondary processes

Advanced processing; EAF + AOD – EAF + VOD

(electric arc furnace + argon oxygen decarburization - electric arc furnace + vacuum oxygen decarbur.)

Degassing methods to produce high quality steels.

After we melt the scraps, C may be in excess then what's needed → we use some argon (neutral gas) to protect the environment into the converter. During decarburization, we may extract also some gas (O_2, CO, CO_2) trapped into the liquid → deoxidization.

EAF's charge:

- Stainless steel scraps compatible with the grade of steel to be produced;
- Plain carbon steel scraps;
- Fe-master-alloys with high content of C, Cr or Ni);

Liquid content (C=1.5-2.5%) to be transferred to either AOD or VOD converter

First, prior decarburizing down to 0.5% (or 0.8%) in the EAF, to reduce subsequent **ladle post-processing** times (purification with **synthetic slags**).

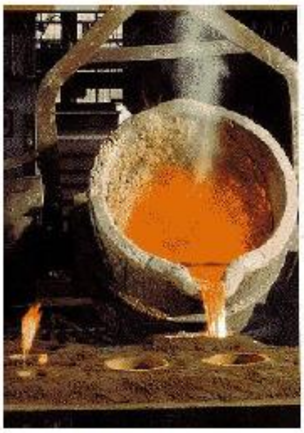
SYNTHETIC SLAGS:

- **Basic slags**: (CaO/MgO-based) => strong reduction of S and P contents in ores
- **Acid slags**: (SiO-based) => severe deoxydation, (block FeS);
- **Acid/basic slag**: (Al2O3/CaO-based) => for both effects.

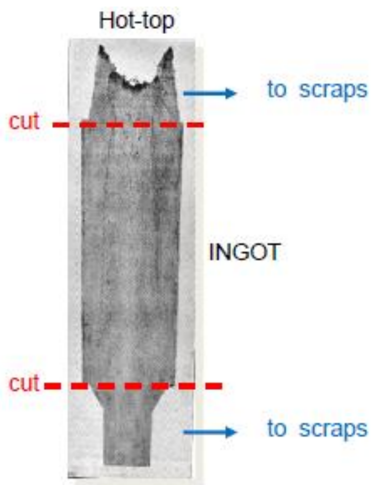
Synthetic slags are used in order to protect the liquid, which should not interact directly with the surface.

VOD: if all the liquid just produced after decarburization is filled into a vacuum station, after create vacuum, it will suck the gas trapped inside the liquid.

Production of cast iron castings



The liquid is poured into sand moulds to produce either large size ingots or complex-shaped castings

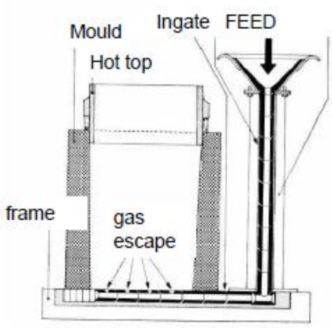


Solidified ingot

Alloying just before casting: once the liquid in contained int the ladle, we put some iron alloys. After melting, there will be an arrangement of the chemical composition.

Sand has very low thermal conductivity, therefore it's used in molds to make the solidification very slow.

Production of Ingots

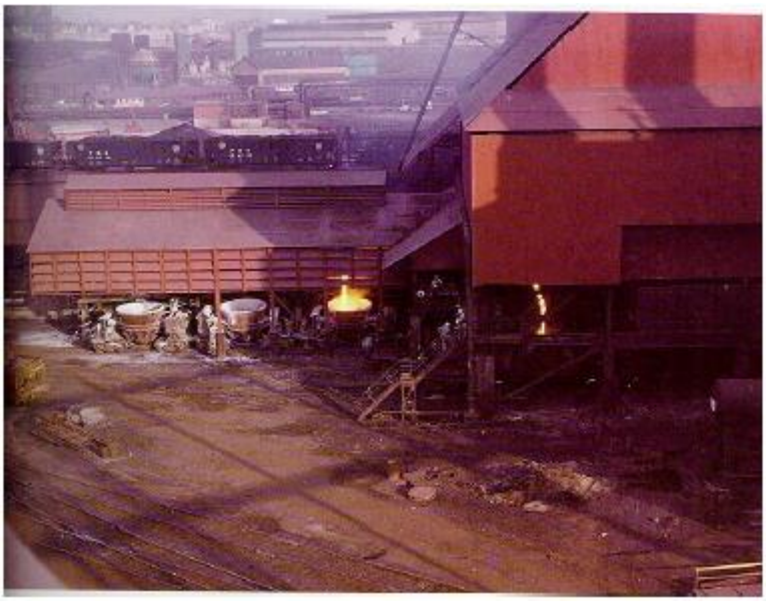


Pouring from bottom

Pouring the molten metal from the top through a gathering system and the ingot is filled from the bottom → in this way we avoid defects on surface and other defects.

The metal is Fe and the impurity is C. Solidification of Fe-C alloy: first solidification of Fe and accumulation of C in the middle → all the carbon and the other residual elements may collect in the center of the ingot after solidification. In order to avoid that, a **hot top** is used on top of the mold; it is made of ceramic which is insulating. Impurities are trapped in the hot top instead of the mold → easy removal of impurities from the metal by cutting the upper part of the ingot (sometimes also the lower part).

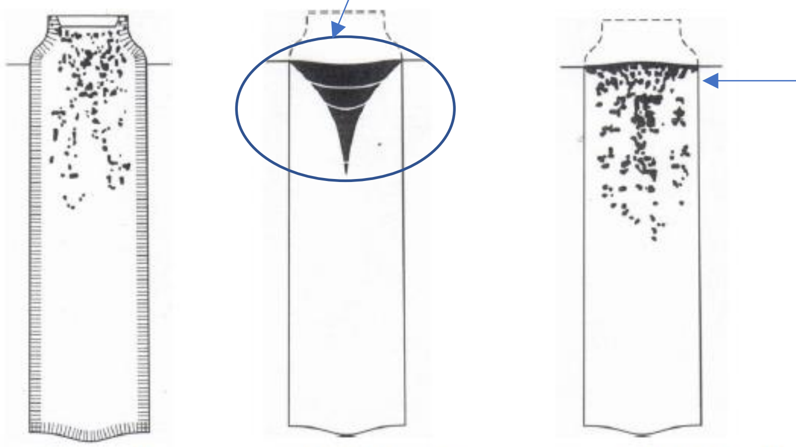
Ladles: melt pouring into ingot



Gas trapping/inclusion control in steels ingots

Casting produces some inclusions due to combination of Al with O_2

Heterogeneous alumina which collects in the top part as last part of the ingot to be solidified



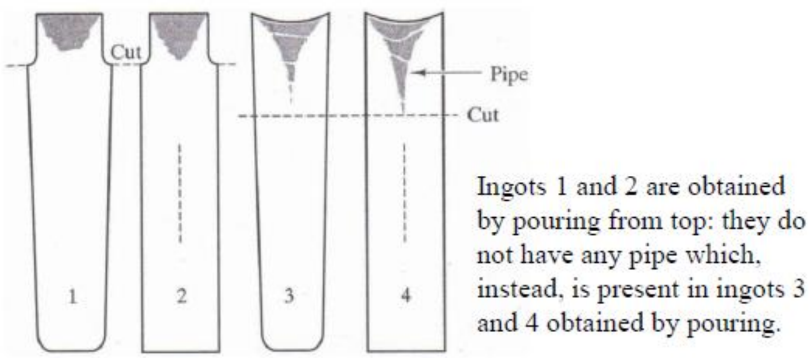
(rimmed steel) (Al-killed steel) (Si-semikilled steel)

Low production of porosity and production of inclusions

Rimmed steel: high porosity content produced by trapped gas.
Al-killed steel: used to combine with O_2 instead of O_2 causing porosity due to CO_2. It is very dangerous when there are ceramic particles.
Si-semikilled steel: less severe than Al but it is not as good to get rid of porosity.

Hot-topping effect on pipe formation

-Ingot (truncated cone to enable ingot extraction from the mold)
-Head (coated with insulating ceramics to retard solidification
-Materozza (contains the pipe to be eliminated)

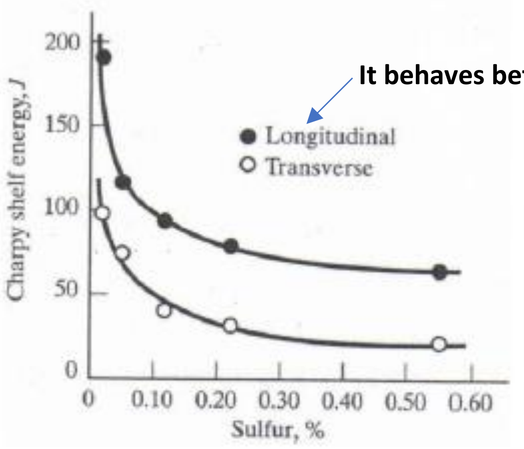


1: best one, hot top + conical shape → better solidification and easier extraction.
2: not tapered shape → solidification with production of segregations in the core.
3: no hot top → macro-segregation. The ingot is cut at a lower level → more scraps.
4: the worst, not tapered, no hot top.

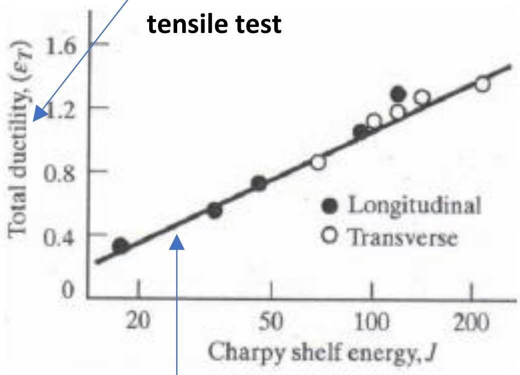
Inclusions: shape control to enhance resilience and ductility

How inclusions affect mechanical properties (they decrease them).

It behaves better



Measured by tensile test



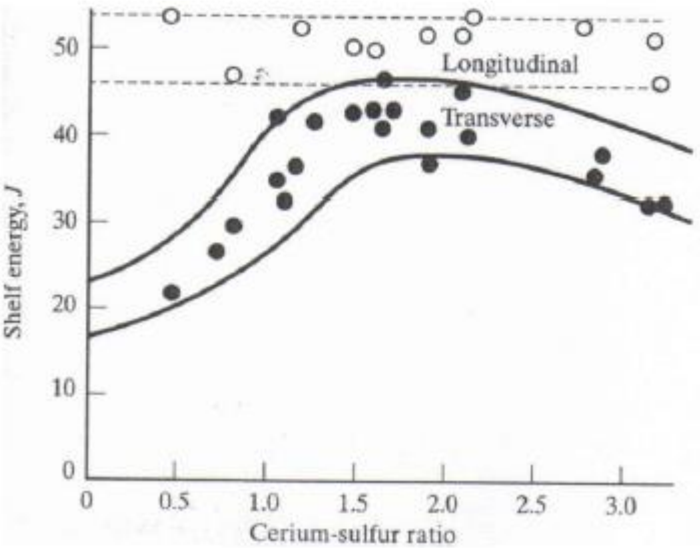
No distinction between longitudinal and transversal. Charpy test: capability of the material to absorb energy. The higher the energy absorbed, the higher the toughness.

Inclusion effects on the upper energy threshold curve obtained by impact test (CVN sample) and on total ductility.

Inclusions may come from:

- **refractories** used to cover the furnace and the ladle before pouring the liquid. The refractory residuals may be trapped in the castings as inclusions;
- **the addition of Al, Si and Ti** (the most severe, as it's highly reactive to oxygen). This leads to the formation of oxides that form small particles which make the metal harder;
- **Intentional inclusions**: to prevent the formation of FeS, we add an extra amount of Mn which leads to the formation of **MnS inclusions**. These are more tolerable than FeS.

Effect of rare earth elements on resilience and ductility after inclusions' shape change



Addition of **Cerium** on the energy measured on transversal direction specimen; after Ce addition, inclusions are spheroidal → isotropic alloy.

The longitudinal direction has higher energy absorption even after we add Cerium. However, with Cerium the energy is quite constant as Cerium-Sulphur ratio increases with transversal, we have a maximum → we don't need a lot of Cerium to make spheroidization of inclusions: only about 1.7%.

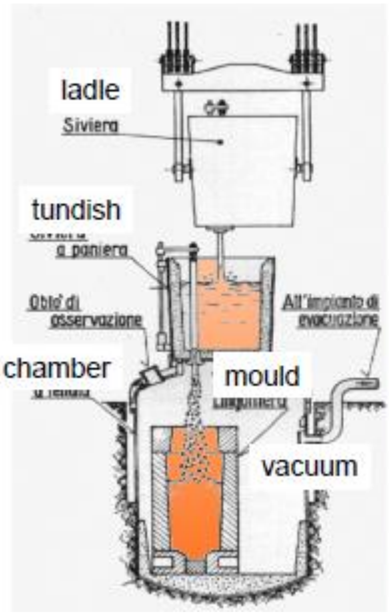
Factors contributing to improve the quality and cost of modern steels
- Usage of Oxygen rather than air (in the past)
- Usage of vacuum processing (to extract low levels of gases trapped in the liquid)
- Usage of inert gases to avoid interaction between hot metal and air
- Usage of mechanical and electromagnetic stirring to homogenize the melt
- Reduced amount of expensive element additions in favor of less expensive elements or improved control of microstructure
- Usage of controlling sensors and systems during processing
- Improvement of basic fundamental knowledge behind processing, steel melt treatment and metallurgy.

Purification of steels
Composition refinement processes are essentially oxydizing, even if countermeasures are taken to minimize additions of O in the decarburized melt pool, a certain amount will remain thereby causing detrimental effects on the steel properties.
Even 100 ppm of O or N are sufficient to induce **embrittlement**.
A **deoxydization operation** may eventually be combined with a decarburizing operation which can be made by adding **deoxidizing alloying**

- Cast iron
- Fe-based alloy with Si, Mn, (Cr, expensive)
- -Al-Si alloys, Al

In these alloys some refinement element, having high chemical affinity to O such as (decreasing order) Al, Si, Mn.
C in Fe-based alloys acts as deoxydizing element. Al may block N and O thereby forming AlN and Al_2O_3 respectively.

Vacuum Degassing



DEGASSING: gases (O_2, N_2, H_2, CO) are dissolved into the liquid and may play a very detrimental role on mechanical properties. The most dangerous is H_2: it causes irreversible embrittlement of the steel ingot.
It is proved that H becomes dangerous when its content is larger than critical values which depend on the steel grade, its distribution in the melt, and amount of impurities segregations among these As, S, etc. and, finally by the cooling rate during forging and subsequent heat treatment.
To reduce/eliminate the absorption of H vacuum degassing is used.
Degassing systems: purification and addition of alloying elements.
Constant pouring of liquid from the bottom part of the tundish → uniform filling of the bottom ingot.
During pouring the mold is kept into vacuum, which extract the gases trapped inside of the liquid and a purified liquid will pour into the ingot and solidify.

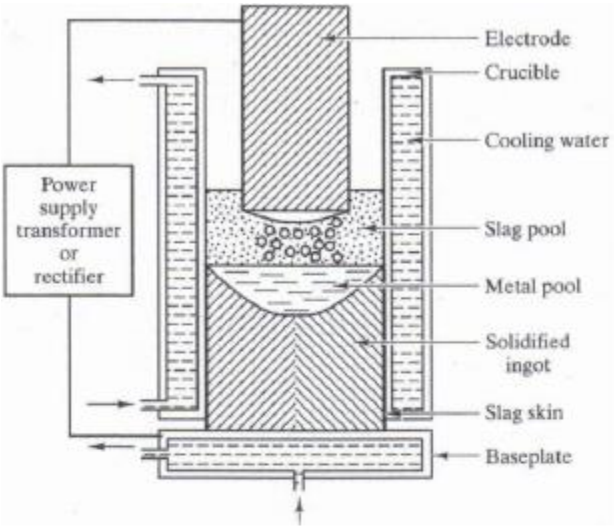
Enhancing Ingot's purity: ESR method

To increase the degree of purity the 'impure' ingot is located at the top of a fused slag mainly composed of **Ba and CaO salts of density approximately of 4 g/cm3** (low density).

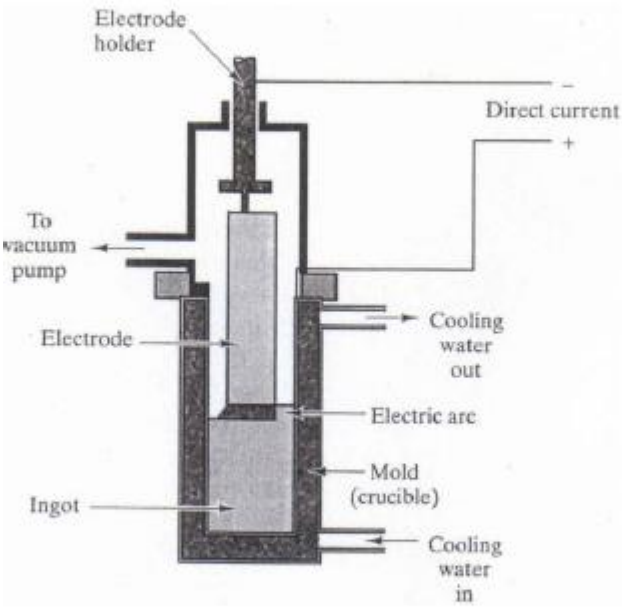
A coil inductor crossed by currents is located at the bottom of the ingot to enable its end remelting such that the small liquid drops, including impurities, cross the porous hot reacting slag.
Steel 'impurities' tend to float above the molten pool whereas the liquid metal tends to collect to the bottom thereby making the new 'clean' ingot after it solidifies.
The steel so obtained is said to be ESR-purified (***Electro Slag Remelting*** – and it is used for the high-quality mechanical industry and in the aerospace sector.

ESR (ElectroSlag Remelting) method

The ingot is remelt while the liquid is covered with a synthetic slag to prevent further oxydization and to purify the molten metal.



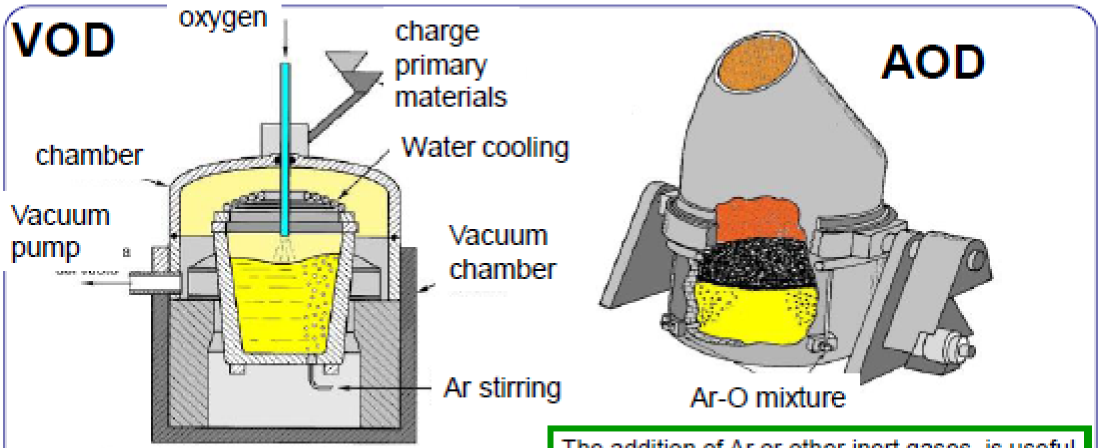
VAR (Vacuum Arc Remelting)



Vacuum remelting with consumable electrode.
A consumable electrode with a predefined composition is remelted by a direct current electric arc in a water-cooled copper crucible. The refining takes place in the arc zone between the electrode tip and the molten metal pool, on top of the ingot which solidifies continuously from the bottom upwards. The controlled solidification of the ingot in the water-cooled copper crucible leads to a sound and homogeneous material with excellent mechanical properties.
In VAR, there's no need for slag.

Production of stainless steel

VOD and AOD are high quality melting methods. VOD is more expensive.

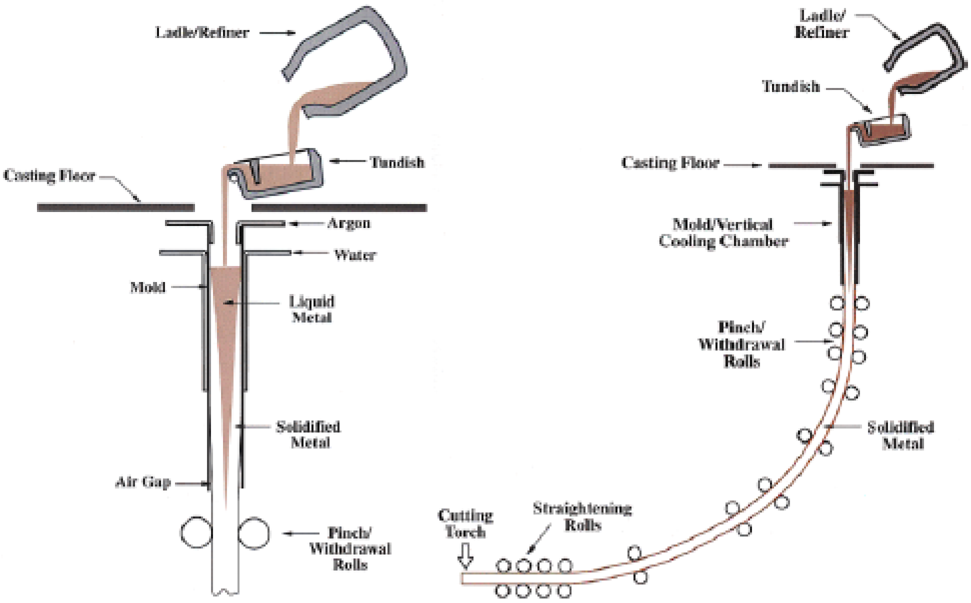


VOD is more suitable to mixed steel making shops in which the rate of produced stainless steel is less than other steels. It is also suitable as simply degassing system of conventional steels

The addition of Ar or other inert gases is useful to dilute CO which develops in the melt pool as a consequence of the exothermal reation between C and O. By varying the Ar/O ratio it is possible to control the partial pressure of CO hence promote the melt pool decarburization

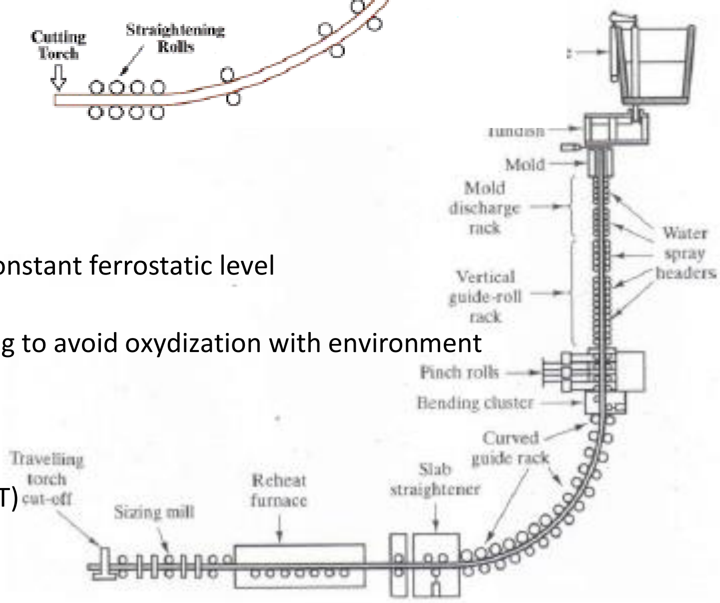
Continuous casting

We have a larger production compared to the discontinuous casting (batch production).



Continuous casting plant

- Ladle capacity: 200-300 t
- Tundish, intermediate ladle to ensure constant ferrostatic level
- Oscillating mold cooled by water
- The liquid is protected by a synthetic slag to avoid oxydization with environment
- Water spray to rapidly cool superficially
- Curved region of the cooled slab
- Straightening of the slab
- Heating furnace (to achieve plastic def. T)
- Rolling strands and slab cutting



Problems

• The presence of slag products (**CaO, SiO2, Al2O3, MgO, Fe2O3**) in the melt pool will inevitably originate undesirable inclusions.

• Inclusions are non metallic phases formed into the metallic matrix.

• **They cause a dramatic drop of the toughness**.

• To avoid this the amount of inclusions has to be limited.

• Solidification of steel ingots is extremely slow and the slag (density about 2 g/cm3) will tend to separate in the melt (density of about 7 g/cm3).

• In continuous casting la cooling rate is much higher hence the steel quality is fair.

Defects after solidification: segregation in I section bars

Fig. 39. Concentric pattern of segregation.
(Courtesy of Titanium Alloy Manufacturing Co.)

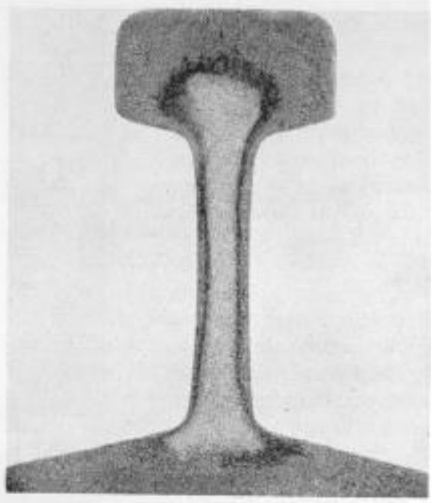
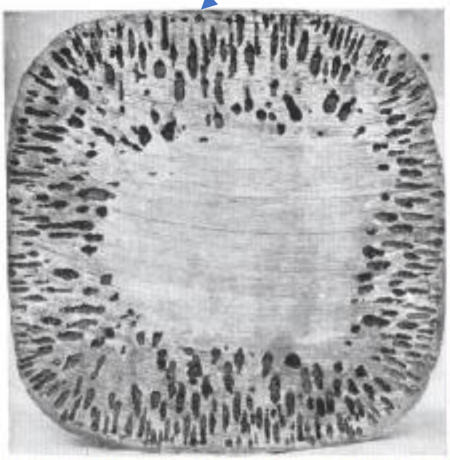


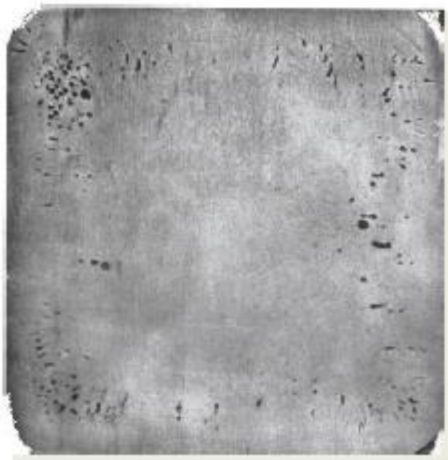
Fig. 40. Concentric pattern of segregation. ½ nat.
(From a paper by A. Herrero and M. de Zubiria. Journ. Iron & Steel Inst., vol. CXVIII, p. 2, 1928.

Defects: blow hole

Porosity



Blow holes in imperfectly killed steel (Cogne, 200x)



Porosity is closed

Analogous blow holes , after rolling (Cogne, 200x).

Inclusions

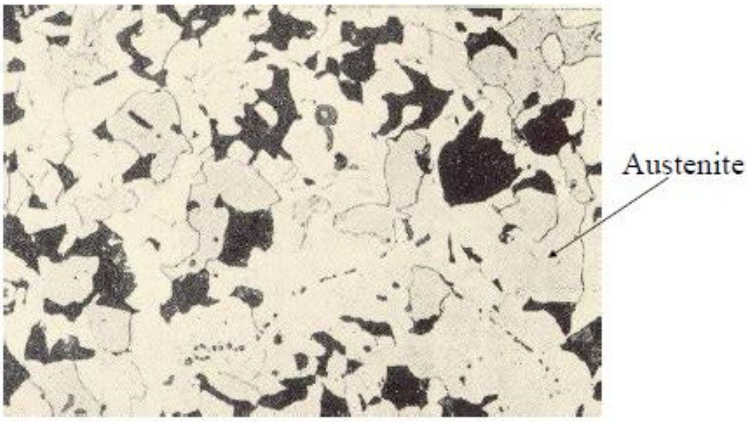
Fracture area of a tensile test sample taken from a transversal direction.
The cross section contains a **large fragmented inclusion**, mostly removed from its original site (light grey).
It has been the cause of premature fracture during tensile test (note the undesired brittle fracture surface).
It's brittle because of the presence of inclusions.

Inclusions (killing → oxides)

Allumina (Cogne, 200x)

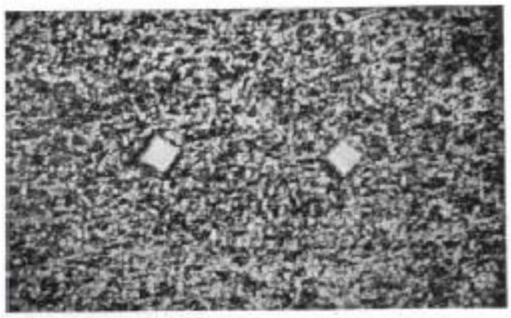
Oxides originated from killing.
In order to improve the refinement due to the presence of oxygen, we have to add Al, which combines with O_2 to form oxides.

Inclusions: oxides at grain boundaries



Oxydes inclusions ariound austenite grain boundaries.

Inclusions (killing → nitrides)



Nitrides. (Cogne, 200x).

Titanium nitrides:

- yellow-pink color-
- non-plastic a the temperature of Hot working
-non-echable. (SIAS 500x)

Inclusions: MnS

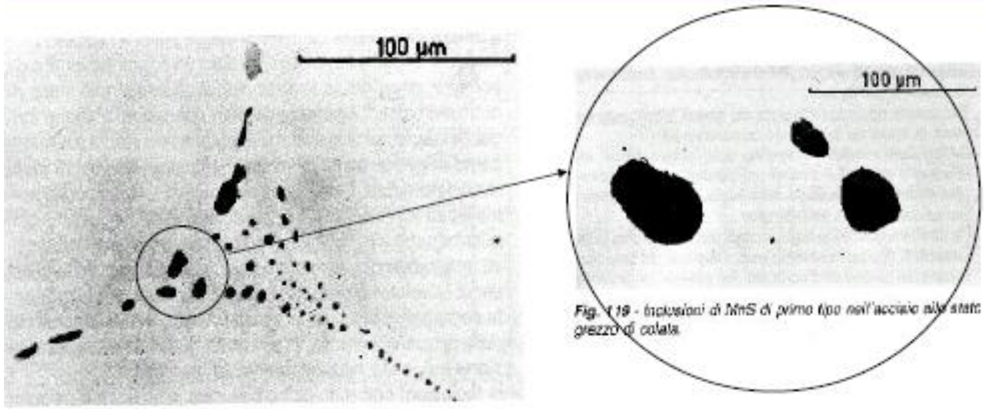


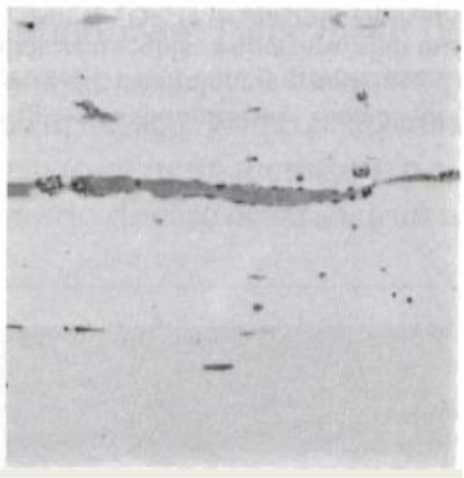
Fig. 119 - Inclusioni di MnS di primo tipo nell'acciaio allo stato grezzo di colata.

Fig. 120 - Inclusioni di MnS di secondo tipo nell'acciaio allo stato grezzo di colata.

Ingot after pouring

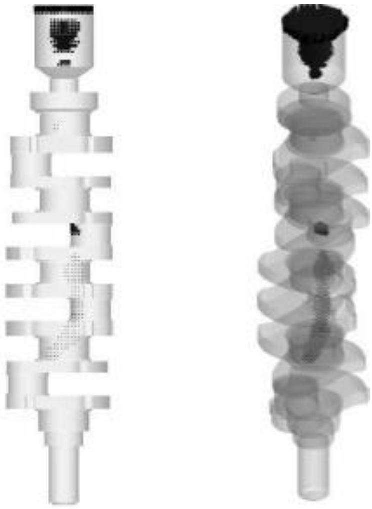
Inclusions: Sulfides (MnS)

- Gray ligh-yello color.
- Very plastic during hot forming, typically MnS;
- Etachable (they are removable) by boiling etching solution of sodium pychrate for 5-10 min



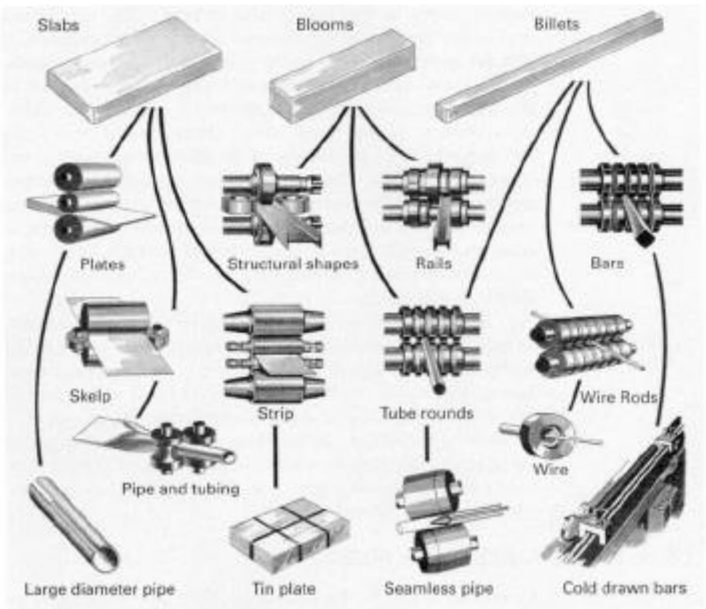
(500x).

Casting defects as detected by X-Ray

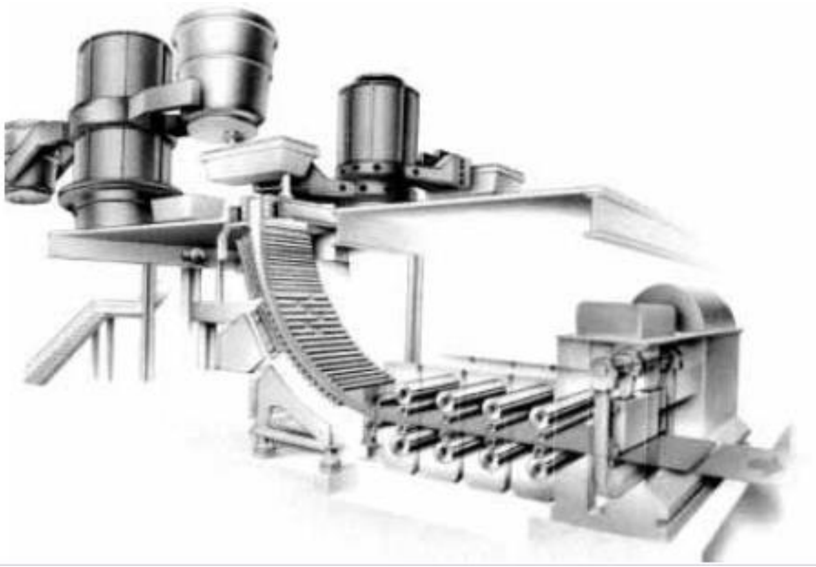


CASTTech

Finished rolled bars



3D continuous casting



Lecture 12

Strengthening of steels by Heat Treatment (phase transformations)

By heat treatment, we can find a good compromise between strength and toughness.
Rapid quenching → martensite + residual stresses.
Slow cooling leads to stabilization.

Heat treatment of steels

On heating

Pre-heating (avoid thermal gradient on heating) (large size and complex geometry pieces)
Annealing (heating metal and allowing it to cool slowly in order to remove internal stress and toughen it)

- isothermal (max softening)
- stabilization (avoid precipitation or grain growth)
- cold work (stress relieving, enhance workability)
- homogenizing or solutionizing (dissolving homogenously the alloying elements in the microstructure, homogeneous microstructure)
- spheroidizing (referring to pearlite or carbides) (refine structure, max toughness in hypereutectoid steels, no phase change)
- recrystallization (refine microstructure, max softening, no phase change)

Austenitization (phase transformation, max softening, max solubility)

On cooling (with phase transformation)

Quenching (rapid cooling, very hard phase)

- hardening
- intermittent (alternating cooling and heating)

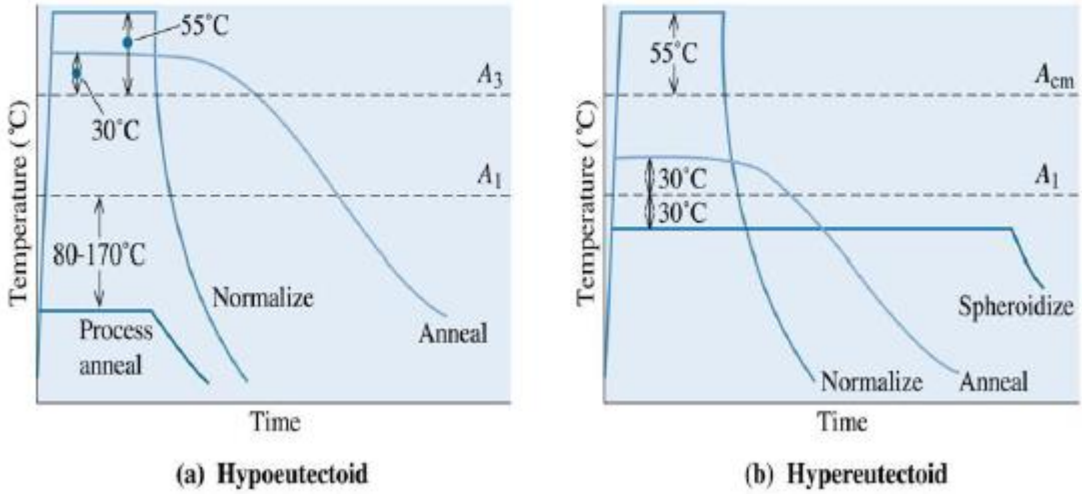
Cycling = heating + cooling (with phase transformations)

- **Normalizing**: heating + air cooling (low expensive refinement, no phase change, relatively fast but not so fast)
- **Tempering** (highest strength):
  - recover (max hardness)
  - softening (max toughness)
  - tempering (optimum compromise between strength/hardness and toughness)
- **Ageing** (not very effective, steel with limit solubility curve): solutionizing + quenching + ageing (maximum strength/hardness)

Isothermal hardening (with phase transformations)

- **austempering** or bainitic
- **martempering (**reduce residual stresses in large pieces)
- **Ausforming** (forming by plastic deformation and hardening) (very used to shape material at high temperature)

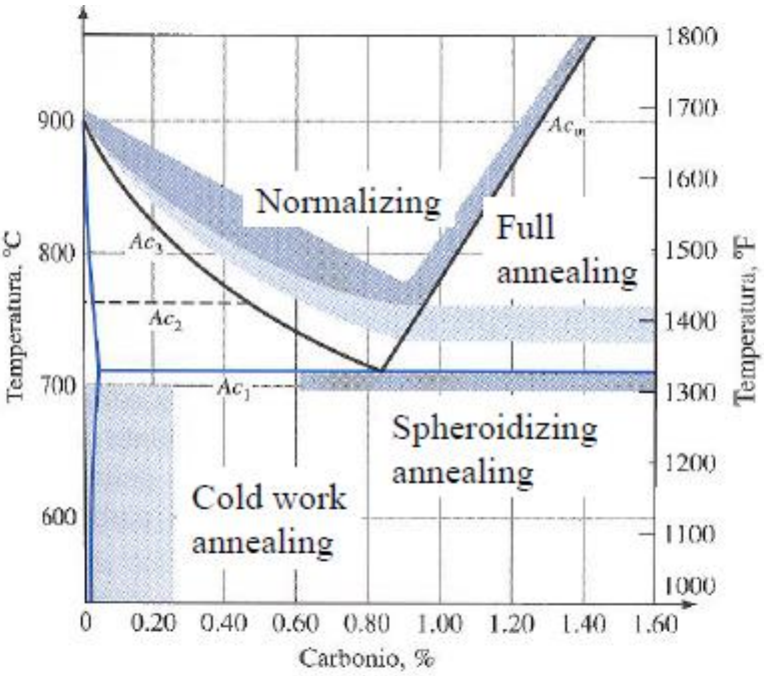
Heat treatments in hypo- and hyper-eutectoid steels



Hypo-eutectoid: high temperature and long process (high energy). We use annealing whenever we want.
Hyper-eutectoid:

- A_{cm}: critical point for cementite
- A_1: normalization is the same as hypereutectoid, but annealing happens at low temperature and long time
- Spheroidize: temperature slightly below A1 and it requires time → expensive, it requires furnace

Temperature Range for Various Heat Treatment of Carbon Steels

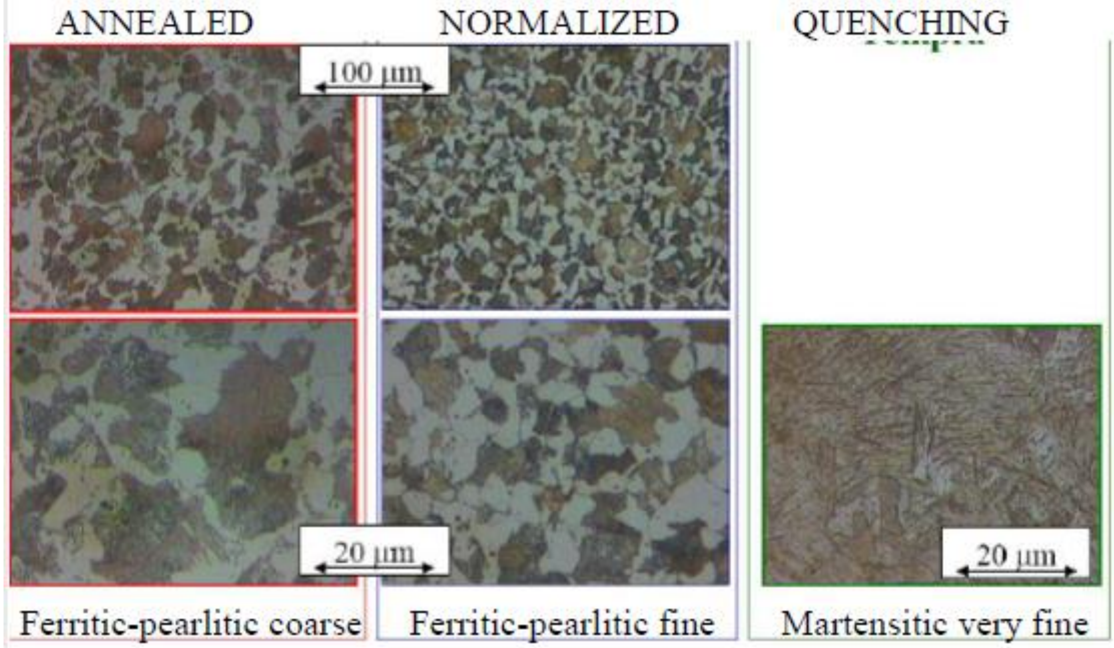


Normalizing → almost 80% above A_3

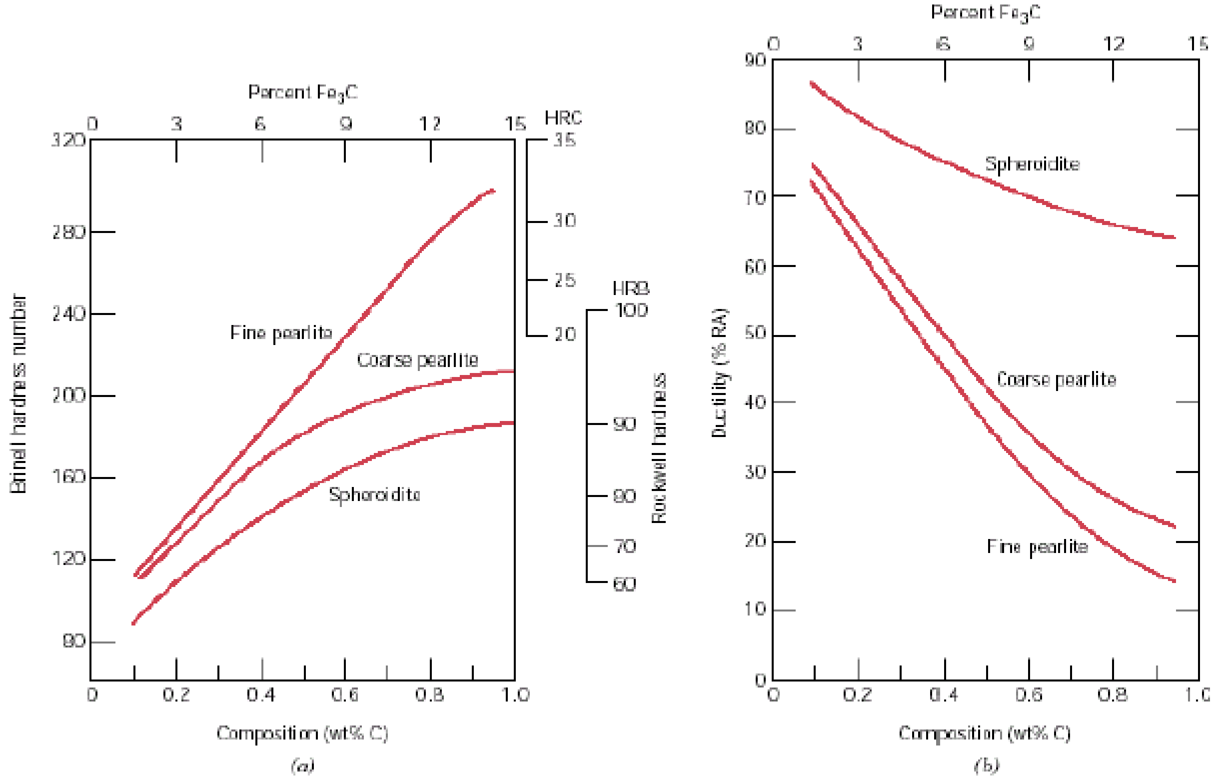
Annealing → temperature below A_1

Spheroidizing → a little bit below A_1; altering heating and cooling → time shortened.

Example of Microstructure after Heat Treatment: AISI1045



Mechanical Properties after Annealing, Normalizing, Spheroidizing in plain Carbon Steels



C controls the hardness of pearlite and also of martensite. Above 1% C there is no hardness, brittleness.
Spheroidite: carbides into soft matrix, like ferrite → low hardness

Phase Transformations: Topics

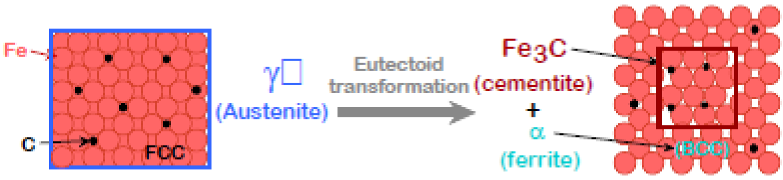
- Kinetics of Phase Transformation
    - Nucleation: homogeneous, heterogeneous
    - Free Energy, Growth
    - Transformation rate

- Isothermal Transformations (TTT diagrams)
- Pearlite, Martensite, Spheroidite, Bainite
- Continuous Cooling (CTT diagrams)
- Mechanical properties

Phase Transformations by heat treatments

Issues to address:

- Transforming one phase into another takes time (kinetics)



Example of solid-to-solid phase transformation

- How does the rate of transformation depend on time and T?
- How can we slow down the transformation so that we can engineer non-equilibrium structures?
- Are the mechanical properties of non-equilibrium structures better?

Phase Transformations: types

Phase transformations – change in the number/character of phases.

- **Simple diffusion-dependent phase transformations**
  - No change in # of phases
  - No change in composition
  - Example: solidification of a pure metal, allotropic transformation, recrystallization,grain growth
- **More complicated diffusion-dependent phase transformations**
  - Change in # of phases
  - Change in composition
  - Example: eutectoid reaction, ferrite, austenite, bainite (microconstituent)
- **Diffusionless phase transformations**
  - Example: highly metastable microconstituent - martensite

Phase transformations

- Most phase transformations begin with the formation of numerous small nucleii (phase nucleation) of the new phase that increase in volume (phase growth) until the transformation is complete.
- Nucleation is the process whereby nuclei (seeds) act as templates for crystal growth.
- Homogeneous nucleation - nuclei form uniformly throughout the parent phase; requires considerable supercooling (typically -ΔT=80-300 °C) or high cooling rates (-ΔT/Δt>100 K/s)
- Heterogeneous nucleation - form at microstructural inhomogeneities (mould walls, impurities, grain boundaries, dislocations, any lattice defect); this is much easier in liquid phase since stable “nucleating surface” is already present; requires slight supercooling (0.1-10 °C).

GBs are more likely to be the starting point for new phases when energy given to the transformation is low because GBs have their own energy. if we put a lot of energy from outside to make cooling so fast, transformation will start in the core at any point within the grains. Any defect into the material will provide the energy for a new phase to be nucleated and grow (heterogeneous nucleation).

Material from liquid to solid cool down at a million rate → it becomes amorphous: all the crystals that will form will be just the points of the alloy in the crystal. If the alloy is made of a matrix (Fe) and an impurity (C), C will be the point where the new crystal will start.
The cooling rate is so fast that no grains are formed and the liquid will be frozen → it becomes amorphous, no time for crystals to form.
To build the crystal, we need diffusion which needs time.

Supercooling

- During the cooling of a liquid, solidification (nucleation) will begin only after the temperature has been lowered below the equilibrium solidification (or melting) temperature Tm. This phenomenon is termed supercooling (or undercooling)
- The driving force for nucleation increases as ΔT increases
- Small supercooling → low nucleation rate - few nuclei - large crystals
- Large supercooling → high nucleation rate – many nuclei - small crystals

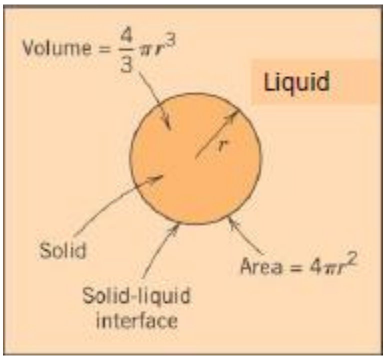
ΔT = transformation point (melting point) is the temperature at which we transform and it should be negative.

[Heat treatment by the induction hardening → we exploit Eddy Currents to heat the material. The process is quite fast (less than one second) → no possibility for diffusion → microstructure remains almost the same.

- Ferrite → austenite (at high-temperature → same grains but with γ (austenization))
- Pearlite → austenite + content in C of pearlite (C will diffuse at high temperature in longer processes (homogeneous))

With fast processes like induction, there is no time for the diffusion → high content of C in some regions → not homogeneous. When the material is quenched, $\gamma \rightarrow \alpha$ with the same martensite (low percentage of C).]

Nucleation of a spherical solid particle in a liquid



- The change in free energy ΔG (a function of the internal energy and enthalpy of the system) must be negative for a transformation to occur.
- Assume that nuclei of the solid phase form in the interior of the liquid as atoms cluster together-similar to the packing in the solid phase.
- Also, each nucleus is spherical and has a radius r.
- Free energy changes as a result of a transformation:
  1) the energy difference between the solid and liquid phases (volume free energy, ΔG_V); and
  2) the solid-liquid phase boundary (surface free energy, ΔG_S).

$$\Delta G_{tot} = \Delta G_s + \Delta G_V$$

Spherical nucleus → it has a volume and a surface which change during the solidification → variation of energy.